

AELP

Adaptive English Learning Platform

Pitch Content & Source Brief for NotebookLM

Core positioning: AELP is an adaptive English learning platform that assesses learners across Listening, Reading, Writing, and Speaking, identifies multiple weak areas, and turns those results into a personalized learning journey.

Use this document as the primary source for generating a product-pitch presentation. Do not invent traction, user counts, accuracy percentages, revenue, or market statistics.

1. The Problem

- English learners may know they need improvement without knowing which skills or sub-skills are causing the problem.
- Assessment and learning are often separated: a learner receives a score and then chooses what to study.
- A single overall English score can hide large differences between Reading, Writing, Listening, and Speaking.
- Generic learning paths can recommend content that does not match a learner's actual weaknesses.
- AELP connects assessment directly to personalized learning recommendations.

2. The AELP Solution

- Core loop: Assess → Identify Weaknesses → Personalize → Learn → Track Progress → Reassess.
- Assess four skills: Reading, Writing, Listening, Speaking.
- Detect weaknesses at skill and sub-skill level.
- Allow a learner to belong to multiple weakness groups.
- Recommend lessons based on actual assessment results.
- Explain why each recommendation is shown.

3. What Makes AELP Distinct

- Assessment-to-learning connection: assessment results influence the next learning action.
- Multi-skill personalization: learners can have multiple simultaneous weaknesses.
- LRWS coverage: the four core English skills are treated separately.
- AI where useful: AI supports Writing/Speaking evaluation and coaching, while teacher-provided answer keys remain authoritative for objective Reading/Listening questions.
- Explainable recommendations: the learner can understand why a lesson was selected.
- Progressive unlocking: the product experience expands as the learner completes assessments.

4. User Journey

- Onboarding → create account and understand the assessment journey.
- Assessment → complete available LRWS assessments.
- Results → view scores, CEFR estimates, strengths, weaknesses, and feedback.
- Weakness mapping → identify priority skills and sub-skills.
- Personalized learning → follow a structured learning journey.
- AI Coach → receive guidance based on actual learner data.
- Progress → track assessment history, lessons, and skill development.
- Reassessment → update the learner profile and recommendations.

5. Assessment Engine

- **Reading:** teacher-provided passage, questions, options, and answer keys; no AI-generated assessment questions; deterministic scoring.
- **Writing:** current example prompt asks the learner to describe a time someone helped them or they helped someone else and how it made them feel; target 120–150 words; evaluated using Groq Llama 3.3 70B Versatile through the backend.
- **Listening:** teacher-provided audio, questions, and answer keys; no AI-generated assessment questions; objective scoring.
- **Speaking:** in-app recording or audio-file upload; Gemini-assisted evaluation of pronunciation, fluency, grammar, vocabulary, coherence, overall speaking performance, and CEFR estimate.

6. CEFR & Results

- Support A1, A2, B1, B2, C1, and C2.
- Reading and Listening use explicit scoring rules and teacher answer keys.
- AI-assessed skills return structured results that are validated and normalized by the backend.
- Maintain one canonical assessment result so Dashboard, Results, Progress, History, and Profile never show conflicting scores.

7. Weakness Detection & Groups

- Identify weaknesses from assessment results.
- Examples: Writing, Speaking, Grammar, Vocabulary, Pronunciation, Fluency, Comprehension, Sentence Structure, Coherence.
- A learner can belong to multiple groups at the same time.
- Weakness groups feed the personalized learning engine.

8. Personalized Learning

- Learning is designed as a journey, not just a flat lesson list.
- Each recommendation should explain why it was selected.
- Example: “Recommended because your writing assessment showed difficulty with sentence structure.”
- Lessons can cover grammar, vocabulary, sentence structure, paragraph organization, pronunciation, fluency, reading comprehension, and listening comprehension.
- Recommendations update as new assessments and lesson progress become available.

9. AI Coach

- Uses the learner's actual assessment results, weaknesses, CEFR profile, and learning progress as context.
- Can answer questions such as “What should I improve?” with guidance tied to the learner's data.
- Must not invent scores or completed lessons.
- Conversation history can be persisted for continuity.

10. Progressive Unlocking

- Before first assessment: assessment-focused experience; no fabricated progress or results.
- After first Reading or Writing assessment: unlock Results, Learning, AI Coach, and Progress.
- After both Reading and Writing: generate a more accurate personalized learning plan.
- After Speaking: unlock the fuller English Profile, including pronunciation and detailed speaking analysis where available.

11. Technology

- Android: Kotlin + Jetpack Compose.
- Backend: FastAPI.
- Primary database: PostgreSQL.
- Database layer: SQLAlchemy/SQLModel + Alembic.
- Networking: Retrofit/OkHttp with typed API models.
- AI: Groq for Writing assessment and Gemini for Speaking assessment, accessed through the backend.
- Security: AI keys and database credentials remain server-side.
- Architecture supports the Android product and the existing AELP web experience.

12. Product Architecture

- Android App → authentication, assessments, audio recording/upload, results, learning, progress, profile, AI Coach.
- FastAPI Backend → authentication, assessment orchestration, scoring, CEFR mapping, weakness detection, personalization, AI integration, APIs.
- PostgreSQL → users, attempts, scores, weaknesses, groups, lessons, recommendations, progress, AI Coach history.
- AI Services → Groq for Writing evaluation; Gemini for Speaking evaluation.

13. Target Users & Use Cases

- Students who want to understand their English level and improve specific weaknesses.
- Learners preparing for academic, professional, or general English improvement.
- Educators/institutions that need structured teacher-controlled assessment content and learner progress data.
- Future institutional use cases can include teacher/admin content management, learner groups, and performance monitoring.

14. Value Proposition & Business Potential

- Learners get a clearer connection between assessment results and what to study next.
- Educators/institutions get structured assessment and learner-performance data.
- The product is designed around personalized improvement rather than a generic course catalog.
- Potential future monetization: learner subscriptions, premium AI coaching, institutional licensing, or educator/admin plans. These are future possibilities, not current traction claims.

15. Roadmap

- Stabilize the Android application and backend.
- Connect production PostgreSQL and secure deployment.
- Finalize LRWS assessment flows.
- Expand teacher/admin content management.
- Expand personalized lesson libraries and recommendation logic.

- Improve AI Coach context and feedback.
- Add richer analytics and long-term progress tracking.
- Explore institutional dashboards and scalable deployment.
- Validate product-market fit through pilot learners and educator/institution feedback.

16. Pitch Narrative

- Opening: “Knowing your English level is not the same as knowing how to improve it.”
- Explain the gap between assessment and personalized action.
- Show that learners can have different weaknesses across LRWS.
- Introduce AELP as the bridge from assessment to adaptive learning.
- Demonstrate assessment → results → weaknesses → personalized lesson.
- Show explainable recommendations and progress.
- Show AI-assisted Writing/Speaking evaluation and teacher-controlled Reading/Listening content.
- Close on the vision: make English improvement more measurable, personalized, and actionable.

17. Suggested Demo Flow

- Open the AELP mobile app.
- Show onboarding/login.
- Open Assessment Hub.
- Demonstrate an assessment.
- Show result + score + CEFR + feedback.
- Show identified weaknesses.
- Open Learning and explain a recommended lesson.
- Show Progress.
- Show English Profile.
- Ask AI Coach what the learner should improve next.

18. Claims to Avoid Without Evidence

- Do not claim user counts, retention, revenue, accuracy percentages, or institutional adoption without verified data.
- Do not claim scientific validation unless supporting evidence is available.
- Do not claim AI scoring is perfectly accurate.
- Do not present roadmap features as already deployed.
- Do not invent competitor rankings or market-share figures.

Closing

AELP is built around a simple idea: **assessment should lead to action.** Instead of stopping at an English score, AELP aims to understand where a learner struggles, turn those weaknesses into a personalized learning journey, and continuously update that journey as the learner

improves.